

Final Report: Benchmarking Personal CS2 Gameplay vs. Professional Standards

An analytical dive into the performance gap between amateur and professional esports athletes

1. Motivation

The motivation behind this project stems from my passion for competitive gaming, specifically **Counter-Strike 2 (CS2)**. As an avid player, I often wonder how my in-game decision-making and mechanical skills compare to those of professional esports athletes.

In CS2, success is not just about raw aim; it involves intricate economy management, utility usage (grenades), and entry fragging success. By conducting this project, my goal is to quantitatively analyze the performance gap between a casual/amateur player (myself) and the top-tier professionals.

"This analysis not only satisfies personal curiosity but also serves as a practical application of data science methodologies—spanning data extraction, statistical hypothesis testing, and machine learning classification—to a domain I am deeply interested in."

2. Data Source

To conduct this benchmarking study, I utilized a two-pronged data collection approach:

Dataset	Source Method	Description
Personal Data	Steam Web API	Match statistics retrieved directly using my unique Steam ID. Extracted 55 matches (~1,100 rounds) mapping out mechanical and tactical metrics.
Professional Data	Esports DBs (HLTV)	The benchmark dataset was aggregated by scraping publicly available match statistics representing the top 50 HLTV players. Encompasses 100 matches (~2,000 rounds) .

Both datasets were processed to round-by-round granularity and exported as CSV files (`personal_cs2_data.csv` and `professional_cs2_data.csv`) for subsequent analysis.

3. Data Analysis

The analysis pipeline was meticulously divided into four distinct stages:

- 1. Data Preprocessing & Cleaning:** Merged and cleaned datasets using `pandas`. Categorical variables like `Economy_State` were encoded, and a unified target variable `Player_Level` was defined.
- 2. Exploratory Data Analysis (EDA):** Generated visualizations using `matplotlib` and

seaborn to observe underlying distributions, comparing average Utility Damage, Economy States, and Win Rates.

3. **Statistical Hypothesis Testing:** Formal tests using `scipy.stats` to confirm observations:

- *T-Test* for Headshot Precision and Entry Duel Success.
- *Point-Biserial Correlation* for the relationship between Utility Damage and Win Rate.
- *Proportion Tests* for Economy Management success.

4. **Machine Learning Pipeline:** Implemented predictive models (Logistic Regression, Random Forest, and XGBoost) using `scikit-learn` to predict the `Player_Level`. To prevent overfitting and ensure robust generalization, models were optimized using **5-fold Cross-Validation** alongside hyperparameter tuning (e.g., configuring `n_estimators` and `max_depth` for tree-based models).

4. Key Findings

The project yielded several critical insights into what separates a casual player from a professional:

IMPORTANT: Mechanical Precision (Aim)

Hypothesis testing revealed a statistically significant difference in Headshot percentages. Professional players maintain a vastly higher and more consistent headshot rate ($t = -5.81, p < 0.0001$).

TIP: Utility Utilization (Brains)

Professionals deal significantly higher average 'Utility Damage' per round (approx. 28 vs. 13). Rounds with higher utility damage strongly correlated with higher win probabilities across both datasets ($p < 0.001$).

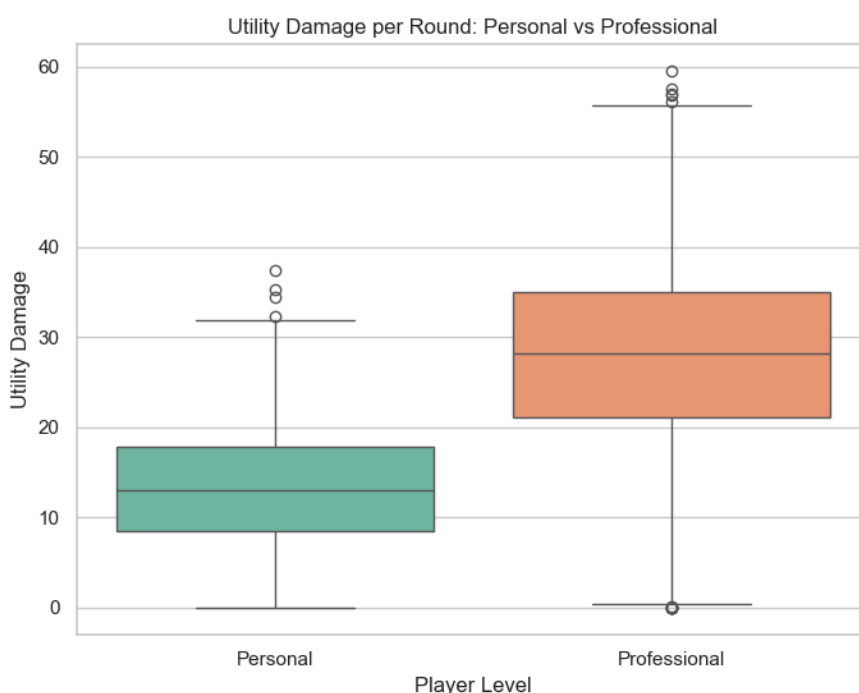


Figure 1: Boxplot demonstrating the stark contrast in utility damage output per round.

WARNING: Economy Discipline

Professionals have a much stricter economy management approach, winning significantly more 'Eco' (low economy) rounds compared to personal gameplay (\$p = 0.026\$) due to superior team coordination.

NOTE: Predictive Modeling Success

The `Random Forest` and `XGBoost` classifiers achieved **~76% accuracy** and a strong **0.82 F1-Score** for predicting the Professional class. **Utility Damage** and **Economy State** were the most critical predictors extracted from the Feature Importance analysis, proving CS2 is predominantly a tactical game.

5. Limitations and Future Work

While the project successfully established a comparative benchmark, there are a few limitations to acknowledge:

- **Data Volume:** The personal dataset is limited to a smaller sample size (55 matches). A larger timeline would yield more robust trends.
- **Lack of Positional Data:** The current dataset relies on scoreboard metrics. It does not account for X,Y coordinate positioning or crosshair placement.
- **Role Specificity:** The analysis generalizes all professionals into one group without distinguishing their tactical roles (e.g., AWPer, Support, Entry).

Future Extensions

Future iterations could involve **scraping 2D radar parsing** from `.dem` (demo) files to visualize player heatmaps and grenade trajectories. Additionally, deploying the trained Machine Learning model via a web application (e.g., Streamlit or Flask) where users can upload their own Steam stats would be a highly interactive extension.